Shoolini University

Progress Report 2 – Development Stage

Submission Date: 20 November 2026

1. Student Details

Name	Md Faiz Alam
Course	BCA (Bachelor of Computer Applications)
Semester	6th Semester
Admission No.	UGO24156735
Project Title	StockFlow – Inventory Management System for Small and Medium Businesses

2. Work Completed

Since the first report, the core of the system has been built module by module on top of the sign-in, layout and product catalogue completed in the initial stage.

Modules developed so far

- Stock Control Engine: every receipt and issue goes through one service function. It locks the product row in PostgreSQL, checks the user's permission and the available quantity, updates the balance and saves a movement record, all in one transaction. Stock can never go negative, and each form carries a unique operation ID so a double click or page refresh is not counted twice.
- Stock Control Page: live totals (active products, units on hand, low stock, out of stock), search and stock-level filters, and Receive / Issue actions for every product.
- Movement History and Corrections: a complete, filterable history showing the time, user, reason, quantity and balance before and after. Mistakes are corrected by a reversal (an exact opposite entry, so the original is never edited) or a stocktake, which adjusts stock to a physical count.
- Supplier Management: supplier contacts, archive/restore, and links between suppliers and the products they provide.
- Purchase Orders: orders are raised to a supplier for a product, quantity and agreed cost. Deliveries are received against the order in parts, with the delivery document number; the page shows the progress and outstanding quantity, and open orders can be cancelled.

- Customers and Sales Orders: customer records and multi-line sales orders that move from Draft → Confirmed → Fulfilled. Fulfilling an order issues stock for every line in one transaction; if any line is short, nothing is issued.
- Reports and CSV Export: current stock report and movement report with search and date filters, page totals, and CSV download (up to 10,000 rows).
- Roles and User Management: three roles – Admin, Manager and Staff – with fixed permissions; a Users & Roles page, a My Account page with password change, and a 15-minute lock after five failed sign-ins.
- Activity Log: records who created, edited or archived records, every stock operation, every order step and every sign-in, including the old and new values of changed fields.

Supporting work

- **Edit-conflict protection:** if two people open the same product, supplier or customer and both save, the second save is refused instead of silently overwriting the first.
- **Demo data command:** one command loads 32 products, 6 suppliers and 256 historical stock movements, plus sample customers and orders, for testing and demonstration.
- **Automated testing:** more than 150 tests covering every module, including tests that run several stock operations at the same time. GitHub Actions now runs all tests on PostgreSQL 17 after every push.

Database at this stage

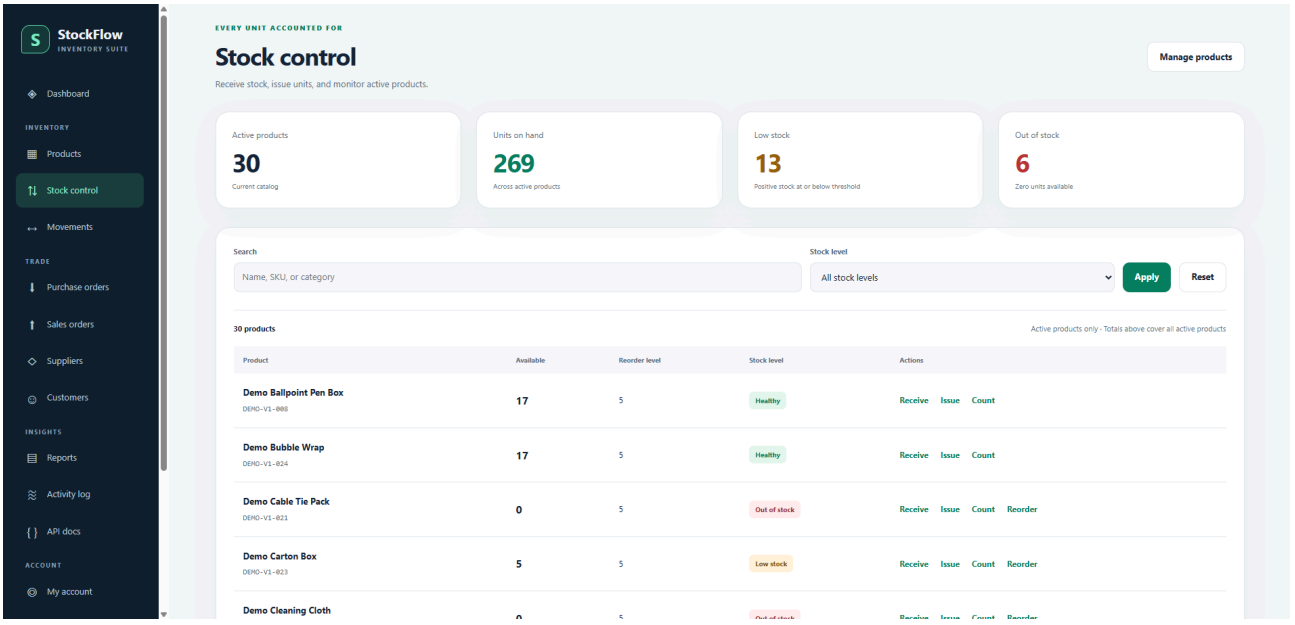
Table	Purpose
users, groups	Accounts and the Admin / Manager / Staff roles
products, stock_balances	Catalogue and current quantity on hand
stock_movements	Every receipt, issue, reversal and stocktake
suppliers	Supplier contacts and product links
purchase_orders	Orders to suppliers and quantity received
customers, sales_orders, sales_order_lines	Customers and their orders
audit_entries	Activity log of user actions

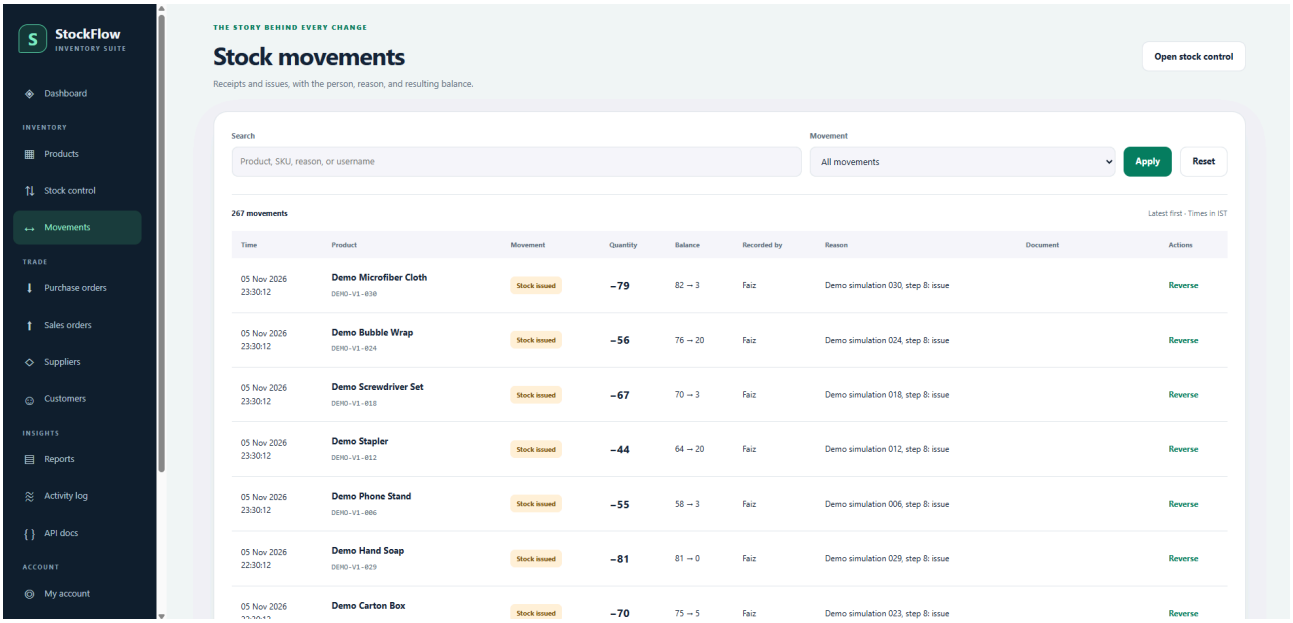
3. System Functionality

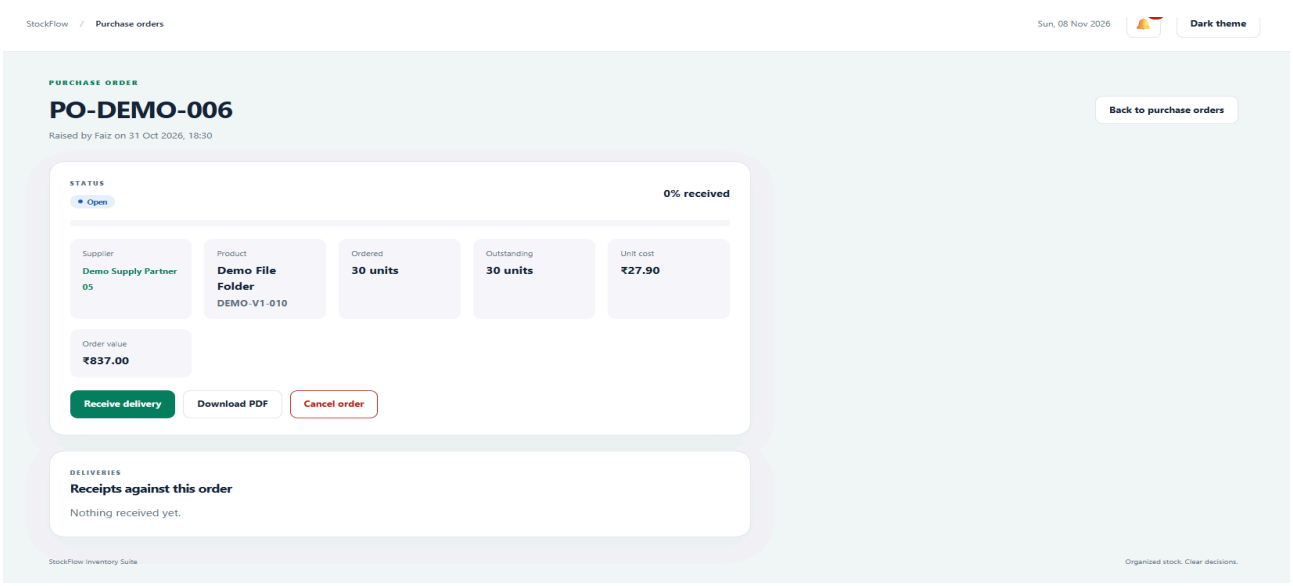
The system currently works as follows:

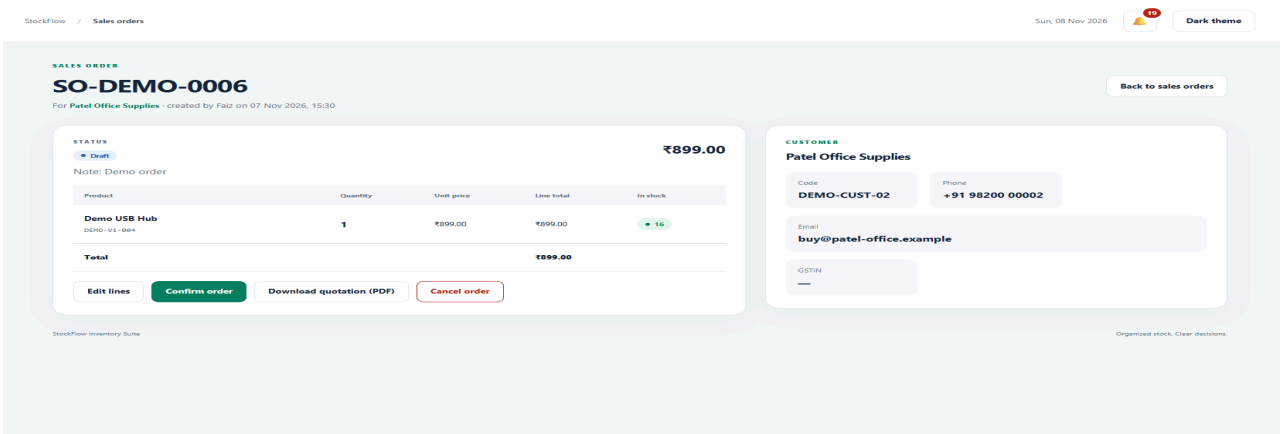
- A user signs in; the menu shows only the pages their role allows.
- A Manager adds products and suppliers and links them together
- When stock runs low, the Manager raises a purchase order to the supplier.
- When goods arrive, Staff record the delivery against the order; stock goes up and the order shows how much is still outstanding.
- A customer order is entered as a sales order, confirmed by a Manager and fulfilled by Staff, which issues the stock for every line.
- Every change is written to the movement history and the activity log, and the reports show current stock and movements for any date range, ready to export as CSV

4. Screenshots / Output




StockFlow
INVENTORY SUITE
Dashboard
INVENTORY
Products
Stock control
Movements
TRADE
Purchase orders
Sales orders
Suppliers
Customers
INSIGHTS
Reports
Activity log
API docs
ACCOUNT
My account
THE STORY BEHIND EVERY CHANGE
Stock movements
Receipts and issues, with the person, reason, and resulting balance.
Open stock control
Search
Product, SKU, reason, or username
Movement
All movements
Apply
Reset
267 movements
Latest first · Times in IST
Time
Product
Movement
Quantity
Balance
Recorded by
Reason
Document
Actions
05 Nov 2026 23:30:12
Demo Microfiber Cloth
DEMO-V1-030
Stock issued
−79
82 → 3
Faiz
Demo simulation 030, step 8: issue
Reverse
05 Nov 2026 23:30:12
Demo Bubble Wrap
DEMO-V1-024
Stock issued
−56
76 → 20
Faiz
Demo simulation 024, step 8: issue
Reverse
05 Nov 2026 23:30:12
Demo Screwdriver Set
DEMO-V1-018
Stock issued
−67
70 → 3
Faiz
Demo simulation 018, step 8: issue
Reverse
05 Nov 2026 23:30:12
Demo Stapler
DEMO-V1-012
Stock issued
−44
64 → 20
Faiz
Demo simulation 012, step 8: issue
Reverse
05 Nov 2026 23:30:12
Demo Phone Stand
DEMO-V1-006
Stock issued
−55
58 → 3
Faiz
Demo simulation 006, step 8: issue
Reverse
05 Nov 2026 22:30:12
Demo Hand Soap
DEMO-V1-029
Stock issued
−81
81 → 0
Faiz
Demo simulation 029, step 8: issue
Reverse
05 Nov 2026
Demo Carton Box
Stock issued
−70
75 → 5
Faiz
Demo simulation 023, step 8: issue
Reverse


StockFlow / Purchase orders
Sun, 08 Nov 2026
Dark theme
PURCHASE ORDER
PO-DEMO-006
Raised by Faiz on 31 Oct 2026, 18:30
Back to purchase orders
STATUS
Open
0% received
Supplier
Demo Supply Partner 05
Product
Demo File Folder
DEMO-V1-010
Ordered
30 units
Outstanding
30 units
Unit cost
₹27.90
Order value
₹837.00
Receive delivery
Download PDF
Cancel order
DELIVERIES
Receipts against this order
Nothing received yet.
StockFlow inventory Suite
Organized stock. Clear decisions.


StockFlow / Sales orders
Sun, 08 Nov 2026
Dark theme
SALES ORDER
SO-DEMO-0006
For Patel Office Supplies · created by Faiz on 07 Nov 2026, 15:30
Back to sales orders
STATUS
Draft
₹899.00
Note: Demo order
Product
Quantity
Unit price
Line total
In stock
Demo USB Hub
DEMO-V1-004
1
₹899.00
₹899.00
16
Total
₹899.00
Edit lines
Confirm order
Download quotation (PDF)
Cancel order
CUSTOMER
Patel Office Supplies
Code
DEMO-CUST-02
Phone
+91 98200 00002
Email
buy@patel-office.example
GSTIN
—
StockFlow inventory Suite
Organized stock. Clear decisions.

5. Challenges Faced

Challenge	How it was solved
Two users updating the same product at the same moment could produce a wrong stock balance.	All stock changes go through one service that locks the product row (SELECT … FOR UPDATE) inside a transaction. Tests run parallel requests to prove it.
A double click or browser refresh submitted the same receipt twice.	Each form carries a unique operation ID; a repeated request returns the original result instead of changing stock again.
Report totals sometimes did not match the rows on the page while stock was changing.	The page rows, row count and totals are now read in a single SQL query using window functions, so they always come from the same moment.
A sales order could be half fulfilled if one product ran out partway.	Fulfilment runs in one transaction with products locked in a fixed order; any shortage rolls back every line.
Two people editing the same record overwrote each other's changes.	Edit forms carry a signed version token; saving an outdated form is refused with a clear message.
Exported CSV files could run formulas when opened in Excel (e.g. a cell starting with "=").	Text that looks like a formula is prefixed with an apostrophe so spreadsheets show it as plain text.

6. Remaining Work

- **Analytics dashboard** with key figures and charts (stock in vs. out, stock value by category, fastestmoving products) and a low-stock alert in the top bar.
- **Live search and filtering** without page reloads (HTMX).
- **PDF documents:** purchase orders, sales invoices and report downloads.
- **REST API** with interactive documentation.
- **Public landing page** and a read-only guest view for visitors.
- **Deployment** on Render (web app) and Neon (PostgreSQL) with production security settings, followed by end-to-end testing on the live site.
- Final report preparation and project zip submission.